

2025EEE3039 MATLAB Coursework

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0.1 Introduction

The aim of this assignment is to facilitate and demonstrate understanding of the two-body problem and torque-free attitude mechanics. There are 8 weeks in total that displays the understanding of Two-body problem and Torque Free attitude mechanics. The coursework is divided into two parts. The first part of the assignment focuses on orbital mechanics and the equations of the two-body problem. It involves the task of integrating the equations of motion in the Earth-Centered Inertial reference frame (ECI) and comparing your solution with the analytical propagations enabled by Kepler's equation along with the derivations of the equations of motion in the Earth-Centered Earth-Fixed reference frame (ECEF) and to convert the initial conditions of the satellite between the ECI and ECEF reference frames. The second part of the coursework deals with attitude kinematics and Euler's equations. The task is to represent the orientation of a satellite with respect to different reference frames and to propagate both Euler's equations and the kinematics equations for a desired time interval. This numerical experiment provides the opportunity to investigate spacecraft orientations and witness the advantages and disadvantages of the different attitude representations that will be introduced throughout the module.

0.1.1 Coursework Data

$$\begin{aligned} \mu = 398600.4418 \text{ km}^3/\text{s}^2 \quad | \quad R_{\oplus} = 6378.137 \text{ km} \quad | \quad \omega_{\oplus} = 7.2921 \times 10^{-5} \text{ rad/s} \\ I_{xx} = 2500 \text{ kg m}^2 \quad | \quad I_{yy} = 5000 \text{ kg m}^2 \quad | \quad I_{zz} = 6500 \text{ kg m}^2 \end{aligned}$$

Part I

A satellite is in a Keplerian orbit around the Earth with initial orbit elements:

$$\alpha = 26561.74 \text{ km} \quad | \quad e = 0.6603 \quad | \quad i = 67.95^\circ \quad | \quad \omega = 321.24^\circ \quad | \quad M_0 = 15.75$$

0.2 Week 2

0.2.1 Kepler's equation using Newton Method:-

- Kepler's Equation: An equation that relates angles and position in an elliptical orbit over time.

$$M = E - e \sin E \quad (1)$$

where, M = Mean Anomaly E = Eccentric Anomaly e = Eccentricity

- Newton's Method: Numerically solves an iterative process and a first derivative. $x_0 = c$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (2)$$

Now solving Kepler's equation using Newton's Method:

Let, $y = f(x_0) + f'(x_0)(x - x_0)$ taking $y = 0 \rightarrow 0 = f(x_0) + f'(x_0)(x_1 - x_0)$
 $x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)}$

Now, $x_0 = c$ and putting in $\rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ Now using equation (1), Let $f(x) = M - E + e \sin E$

and Let, $x = E$ for ease of understanding, and performing the calculations
 $f(E) = 15.75 - E + 0.66603 \sin E$
 $f'(E) = -1 + 0.6603 \cos E$ $E(0) = M$

and then running the program will give the following results:

Note: $|f(E_k)| = tol$

Numerical Values:

$e = 0.6603$

$Tol = 10^{-10}$

Mean anomaly M = 0.2748893572 rad (15.75000000 deg)

True anomaly theta = 1.358802444288 rad (77.85364525 deg)

Kepler and True Anomaly Results:

- (i) Converged after 5 iterations because $|f(E)| < tol$
- (ii) Eccentric Anomaly (E) = 0.70053788 radians

0.2.2 Periapsis or Apoapsis

Using the data,

$a = 26561.74$ km ; $e = 0.6603$

Calculated:

$$E = 40.129^\circ$$

Using the Orbital Equation formula derived in Week2 lecture,

$$r = a(1 - e \cos E) = 26561.74(10.6603 \cos(40.1219)) = 13150.31912 \text{ km} \quad (3)$$

Therefore,

for periapsis, $r_p = a(1 - e)$; for apoapsis, $r_a = a(1 + e)$

RESULT:

• *Orbital radii (km) :*

$$r = 13153.46864 \text{ km (current)}$$

$$r_p = 9023.02308 \text{ km (periapsis)}$$

$$r_a = 44100.45692 \text{ km (apoapsis)}$$

• *Distances :*

$$|r - r_p| = 4130.44557 \text{ km}$$

$$|r_a - r| = 30946.98828 \text{ km}$$

Conclusion: the Satellite is much closer to Periapsis.

0.3 Week 3

Given $coe = [a \ e \ i \ \Omega \ \omega \ \theta]_T$

0.3.1 Orbital Period

Orbital Period,

$$P = 2\pi \sqrt{\frac{a^3}{\mu}} \quad (4)$$

where, a = semi-major axis μ = Earth's gravitational parameter. Equations to be used,

1. Kepler's Equation, $M = E - e \sin E$
2. $\tan\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+e}{1-e}} \tan\left(\frac{E}{2}\right)$
3. $M = M_0 + n(t - t_0)$
4. True Anomaly, $\theta = 2 \cdot \arctan\left(\frac{\sqrt{1+e} \sin(E/2)}{\sqrt{1-e} \cos(E/2)}\right)$

RESULT:

Orbital Period $P = 43080.370 \text{ s}$

Plot:

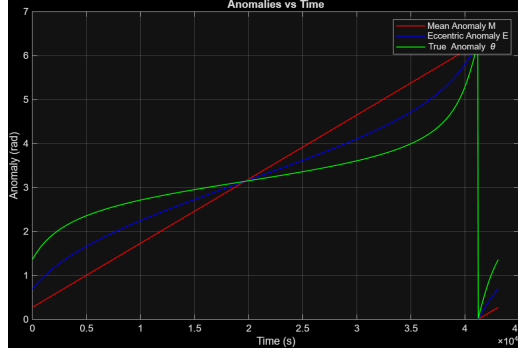


Figure 1: Anomalies vs Time

0.3.2 COE2RV

Following the recipe for transformation from Classical Orbit Elements to Cartesian Coordinates from Week3:

Goal: $\vec{o\vec{e}} = [a \ e \ i \ \Omega \ \omega \ \theta]^T \rightarrow (\vec{r}, \vec{v})$

Methodology: Use properties of Rotation Matrices to:

- Find DCM from the ECI frame to the Perifocal frame and viceversa
- Map position and inertial velocity vectors from P-frame to ECI

$$[PI] = R_3(\Omega)R_1(i)R_3(\omega)$$

$$R_3\Omega =$$

$$\begin{bmatrix} \cos\Omega & \sin\Omega & 0 \\ -\sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$R_1i =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{bmatrix}$$

$$R_3\omega$$

$$\begin{bmatrix} \cos\omega & \sin\omega & 0 \\ -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• *Input* : a = semi-major axis, e = eccentricity, i = inclination, Ω = RAAN, ω = argument of perigee, θ = true anomaly, $\mu = 3.986 \times 10^5 \text{ km}^3/\text{s}^2$

• *Output* : The state vector in the ECI frame is given by: $\mathbf{X}_{ECI} = \begin{bmatrix} \mathbf{r}_{ECI} \\ \mathbf{v}_{ECI} \end{bmatrix}$

The full state vector is:

$$\mathbf{X} = [r_x \ r_y \ r_z \ v_x \ v_y \ v_z]^T$$

From Week 2 lectures, the orbital radius is obtained from the orbit equation:

$$r = \frac{h^2/\mu}{1 + e \cos \theta}$$

or equivalently:

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

The Direction Cosine Matrix (DCM) is:

$$[IP] = R_3(-\Omega) R_1(-i) R_3(-\omega)$$

The position and velocity in the perifocal frame are:

$$\mathbf{r}_{pf} = [r \cos \theta \ r \sin \theta \ 0]^T$$

$$\mathbf{v}_{pf} = \left[-\left(\frac{\mu}{h}\right) \sin \theta \quad \left(\frac{\mu}{h}\right) (\cos \theta + e) \quad 0 \right]^T$$

Transforming to the ECI frame:

$$\mathbf{r}_{ECI} = [IP] \mathbf{r}_{pf}$$

$$\mathbf{v}_{ECI} = [IP] \mathbf{v}_{pf}$$

Thus, the final state vector is:

$$\mathbf{X}_{ECI} = [\mathbf{r}_{ECI} \ \mathbf{v}_{ECI}]^T$$

RESULTS:

- At $t=0$: $M=0.27488936$ rad, $E=0.70053788$ rad, $\theta = 1.35880244$ rad
- At $t=P$: $M=0.27488936$ rad, $E=0.70053788$ rad, $\theta = 1.35880244$ rad
- Final State: For a closed two-body orbit, the satellite returns to the same inertial state after one orbital period because $M \rightarrow (0, 2\pi)$ from $t = 0$ to $t = P$.

Hence,

$$\mathbf{X}_f \approx \mathbf{X}_0$$

Plot

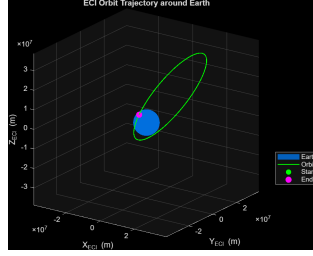


Figure 2: ECI Orbit Trajectory with Earth

0.4 Week 4:

Numerical Integration in the ECI frame

The satellite moves under the gravitational influence of the Earth according to the equations of the two-body problem:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \vec{r} \quad (5)$$

Position vector from Earth's center:

$$\mathbf{r} = [x \ y \ z]^T, \quad \ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r}, \quad \mu = \text{gravitational parameter of Earth}$$

Magnitude:

$$r = \|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}$$

ODE solver in MATLAB takes only first order differentital equations.

$$\dot{\mathbf{X}} = f(t, \mathbf{X})$$

Converting to first order differentital equations: Let:

$$\mathbf{X} = [\mathbf{r} \ \mathbf{v}]^T = [x \ y \ z \ v_x \ v_y \ v_z]^T$$

Component form:

$$\begin{aligned} \dot{x} &= v_x, & \dot{y} &= v_y, & \dot{z} &= v_z \\ \dot{v}_x &= \frac{\mu x}{(x^2 + y^2 + z^2)^{3/2}}, & \dot{v}_y &= \frac{\mu y}{(x^2 + y^2 + z^2)^{3/2}}, & \dot{v}_z &= \frac{\mu z}{(x^2 + y^2 + z^2)^{3/2}} \end{aligned}$$

MATLAB coursework, the system for ODE solvers,

$$[t_{\text{out}}, Y_{\text{out}}] = \text{ode}(t_{\text{span}}, X_0, \text{ode-options}, \text{initial condition}, \text{parameter of function})$$

For MATLAB built-in solvers:

$$[t_{\text{num}}, X_{\text{num}}] = \text{ode45}(@(\mathbf{t}, \mathbf{X}) \text{TBP_ECI}(\mathbf{t}, \mathbf{X}, \mu), t_{\text{span}}, X_0, \text{ode_opts})$$

0.4.1 Comparing Numerical vs Analytical Solution

From Week3 analytical two-body solution:

For each time step t_k :

$$X_{\text{num}}(k, :) = [x(t_k) \quad y(t_k) \quad z(t_k)]$$

for each time t-k:

$$dX/dt = \text{TBP_ECI}(t, X, \mu) = [\dot{\mathbf{r}} \quad \dot{\mathbf{v}}]^T$$

Thus:

$$\frac{d\mathbf{r}}{dt} = \mathbf{v} - \mathbf{k}, \quad \frac{d\mathbf{v}}{dt} = -\frac{\mu}{|\mathbf{r} - \mathbf{k}|^3} \mathbf{r} - \mathbf{k}$$

Error Calculation Once analytical and numerical values are known:

(i) Position Error

- Norm position error at t=0: $0.000000e + 00m$
- Norm position error at t=P: $1.047345e - 01m$
- Max position error over [0,P]: $1.322985e - 01m$

(ii) Velocity Error

- Max velocity error over [0,P]: $7.458732e - 05m/s$

Plot:

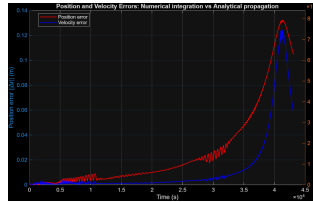


Figure 3: 4.1 Position and Velocity Errors

0.4.2 Specific Angular Momentum

Specific angular momentum of satellite:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}, \quad h = |\mathbf{h}| = \text{constant}$$

To verify wheather the specific angular momentum of a satellite is integral of motion of the two body problem, the following must kept in mind :

Conditions:

1. No numerical integration error example : tolerance, skip size, order of integral.
2. No mismatched units

For Theoretical specific angular momentum,:

$$h = \sqrt{\mu a(1 - e^2)}$$

RESULT: The analytical values achieved at each time interval should nearly constant and equal to the theoretical value $\sqrt{\mu a(1 - e^2)}$ within the numerical error defined in the script.

- Theoretical specific angular momentum $|h_{theo}| = 7.727486e+10(m^2/s)$
- Mean Value of Analytical Specific Angular Momentum $|h_{ana}| = 7.727486e+10(m^2/s)$

Plot

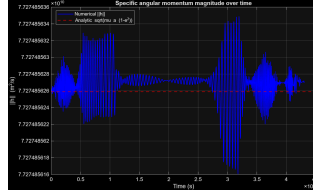


Figure 4: 4.2 Specific angular momentum norm

0.5 Week 5:

0.5.1

Transport theorem: According to Transport Theorem, the derivative of a vector will look different when compared to an inertial frame, if the frame is rotating. Transport theorem: The transport theorem relates time derivatives with respect to two different reference frame: for any vector, the ECI frame and ECEF frame can be related by, $\dot{\mathbf{a}}_I = \dot{\mathbf{a}}_J + \boldsymbol{\omega}_{J/I} * \vec{\mathbf{a}}$

$$\left(\frac{d\mathbf{A}}{dt} \right)_I = \left(\frac{d\mathbf{A}}{dt} \right)_J + \boldsymbol{\omega}_{J/I} \times \mathbf{A}$$

I = inertial frame J = rotating frame (ECEF)

Apply to position, velocity, acceleration:

Velocity transformation: First derivative, $v_I = \frac{dr_I}{dt}$ using, transport theorem, $\frac{dr_I}{dt} = \frac{dr_J}{dt} + \omega_{J/I} * \vec{r}$

$$\mathbf{v}_I = \mathbf{v}_J + \boldsymbol{\omega}_{J/I} \times \mathbf{r}$$

Second Derivative

$$\begin{aligned} \mathbf{a}_I &= \frac{d\mathbf{v}_I}{dt} = \frac{d}{dt} \left(\frac{d}{dt} \left[\frac{d\mathbf{r}}{dt} + \boldsymbol{\omega}_{J/I} \times \mathbf{r} \right] \right) \\ &= \left[\frac{d\mathbf{v}_r}{dt} \right] + \frac{d}{dt} (\boldsymbol{\omega}_{J/I} \times \mathbf{r}) + [\boldsymbol{\omega}_{J/I} \times \mathbf{v}_r] + (\dot{\boldsymbol{\omega}}_{J/I} \times \mathbf{r}) + (\boldsymbol{\omega}_{J/I} \times (\boldsymbol{\omega}_{J/I} \times \mathbf{r})) \end{aligned}$$

Hence,

$$\mathbf{a}_I = \mathbf{a}_r + \dot{\boldsymbol{\omega}}_{J/I} \times \mathbf{r} + 2\boldsymbol{\omega}_{J/I} \times \mathbf{v}_r + \boldsymbol{\omega}_{J/I} \times (\boldsymbol{\omega}_{J/I} \times \mathbf{r})$$

where:

$$2\boldsymbol{\omega}_{J/I} \times \mathbf{v}_r : \text{Coriolis acceleration}$$

$$\boldsymbol{\omega}_{J/I} \times (\boldsymbol{\omega}_{J/I} \times \mathbf{r}) : \text{centrifugal acceleration}$$

*Note: Considering two cases:

(i) **Constant rotation rate** $\dot{\boldsymbol{\omega}}_{J/I} = 0$

Resulting accelerations will be :

- (a) gravitational = $\mathbf{a}_I$
- (b) Coriolis = $2\boldsymbol{\omega}_{J/I} \times \mathbf{v}_r$
- (c) centrifugal = $\boldsymbol{\omega}_{J/I} \times (\boldsymbol{\omega}_{J/I} \times \mathbf{r})$

(ii) **Variable rotation rate** $\dot{\boldsymbol{\omega}}_{J/I} \neq 0$

Resulting acceleration will be:

- (a) gravitational = $\mathbf{a}_I$
- (b) Coriolis = $2\boldsymbol{\omega}_{J/I} \times \mathbf{v}_r$
- (c) Euler = $\dot{\boldsymbol{\omega}}_{J/I} \times \mathbf{r}$
- (d) centrifugal = $\boldsymbol{\omega}_{J/I} \times (\boldsymbol{\omega}_{J/I} \times \mathbf{r})$

0.5.2

$$\mathbf{a}_J = \mathbf{a}_I - 2\boldsymbol{\omega}_{J/I} \times \mathbf{v}_J - \boldsymbol{\omega}_{J/I} \times (\boldsymbol{\omega}_{J/I} \times \vec{r})$$

To write in MATLAB, we need to convert into a first-order system so that the MATLAB ODE solver can accept it:

$$dX/dt = f(t, \mathbf{X})$$

State Vector

$$\mathbf{X} = [\vec{r} \quad \vec{v}]^T$$

$$\vec{r} \text{ (position vector)} = [x \quad y \quad z]^T$$

$$\vec{v}_{\mathbf{F}} \text{ (velocity in rotating frame)} = [v_x \quad v_y \quad v_z]^T$$

$$\boldsymbol{\omega} \text{ (Earth angular velocity vector)} = [0 \quad 0 \quad \omega_{\oplus}]^T$$

To place into MATLAB first-order form:

State vector:

$$\mathbf{X} = [\mathbf{r}_J \quad \mathbf{v}_J]^T = [x \quad y \quad z \quad v_x \quad v_y \quad v_z]^T$$

Thus:

$$\dot{\mathbf{X}} = [\mathbf{v}_J \quad \mathbf{a}_J]^T$$

Earth rotates about Z-axis:

$$\boldsymbol{\omega} = [0 \quad 0 \quad \omega]^T$$

Component ECEF equations (from notebook):

$$\dot{x} = v_x$$

$$\dot{y} = v_y$$

$$\dot{z} = v_z$$

$$\dot{v}_x = -\frac{\mu x}{r^3} - 2(\omega * v_J)_x - (\omega(\omega * \vec{r}))_x$$

$$\dot{v}_y = -\frac{\mu y}{r^3} + 2(\omega * v_J)_y - (\omega(\omega * \vec{r}))_y$$

$$\dot{v}_z = -\frac{\mu z}{r^3} + 2(\omega * v_J)_z - (\omega(\omega * \vec{r}))_z$$

RESULT:

$$\dot{\mathbf{X}} = [\mathbf{v}_f, \mathbf{a}_f]^T$$

0.5.3

$$[FI] = I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

For Position vector: At $t_0 = 0$

$$\mathbf{r}_{ECEF}(t_0) = [FI] \mathbf{r}_{ECI}(t_0)$$

$$r_{ECEF} = r_{ECI} \quad \text{at } t_0$$

Velocity (Transport Theorem):

$$\mathbf{v}_J = \mathbf{v}_I - \boldsymbol{\omega} \times \tilde{\mathbf{r}}_I$$

Initial ECEF state will be:

$$\mathbf{r}_I = \begin{bmatrix} x_0 & y_0 & z_0 \end{bmatrix}^T, \quad \mathbf{v}_I = \begin{bmatrix} v_{x0} & v_{y0} & v_{z0} \end{bmatrix}^T$$

$$\mathbf{v}_J = \mathbf{v}_I - \begin{bmatrix} 0 & 0 & \omega \end{bmatrix}^T \times \mathbf{r}_I$$

RESULT:

Orbital period $P = 43081.990678$ s (11.967220 hours)

Converged after 5 iterations

Eccentric Anomaly (E) = 0.70053788 radians

Initial ECEF state (r [m], v_F [m/s]) :

$$\begin{bmatrix} r & v_F \end{bmatrix}^T = 1.0 \times 10^7 \begin{bmatrix} 0.2885 & -1.0276 & 0.7688 & 0.0002 & 0.0001 & 0.0006 \end{bmatrix}^T$$

Final ECEF state after 10 periods:

$$\begin{bmatrix} r & v_F \end{bmatrix}^T = 1.0 \times 10^7 \begin{bmatrix} 0.2885 & -1.0276 & 0.7688 & 0.0002 & 0.0001 & 0.0006 \end{bmatrix}^T$$

Norm of difference (final - initial) = 6.967296e+01

Plot

Comment: If the orbital period of the satellite is not an integral multiple of the rotation period of the Earth $P_{earth}(86164s)$, the final ECEF state of the satellite will differ from the initial state. The difference in norm shows the difference between initial and final ECEF state.

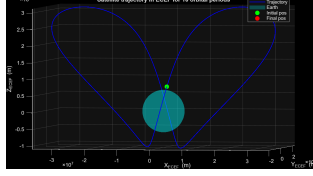


Figure 5: 5.4 ECEF Trajectory

0.6 Week 6: Direction Cosine Matrix

Direction-cosine matrix transforms vector coordinates from one reference frame to another. In other words, the direction-cosine matrix transforms coordinates of a vector expressed in the J -frame to coordinates expressed in the I -frame (or vice versa depending on the convention used).

From the properties of unit vectors and the dot product, the cosine of the angle between the i -th basis vector of one frame and the j -th basis vector of another frame is the dot product of the corresponding unit vectors:

$$\hat{j}_p \cdot \hat{i}_q = \cos(\theta)_{pq}$$

$$\mathbf{B}(\Theta) = \frac{1}{\sin \Theta_2} \begin{bmatrix} \Theta_3 & 0 & 0 \\ \sin \Theta_3 & \Theta_2 \cos \Theta_3 & -\sin \Theta_2 \cos \Theta_3 \\ -\Theta_2 \cos \Theta_3 & \Theta_2 \sin \Theta_3 & \sin \Theta_2 \end{bmatrix}$$

For better understanding of the DCM, I have resourced from multiple sources and hence concludes, For two reference frames, and , we need a third references frame that does not depends on the two reference frames, say . Thus the direction-cosine matrix $\mathbf{C}$ can be assembled from these cosines. For example, writing the matrix in terms of generic angles α_{ij} we get

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}.$$

Interpretation in terms of basis vectors:

If $\{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3\}$ are the basis unit vectors of one frame and $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ are the basis unit vectors of another frame, then each matrix element is the cosine of the angle between a pair of basis vectors:

$$\mathbf{C} = \begin{bmatrix} \cos(\angle(\mathbf{d}_1, \mathbf{e}_1)) & \cos(\angle(\mathbf{d}_1, \mathbf{e}_2)) & \cos(\angle(\mathbf{d}_1, \mathbf{e}_3)) \\ \cos(\angle(\mathbf{d}_2, \mathbf{e}_1)) & \cos(\angle(\mathbf{d}_2, \mathbf{e}_2)) & \cos(\angle(\mathbf{d}_2, \mathbf{e}_3)) \\ \cos(\angle(\mathbf{d}_3, \mathbf{e}_1)) & \cos(\angle(\mathbf{d}_3, \mathbf{e}_2)) & \cos(\angle(\mathbf{d}_3, \mathbf{e}_3)) \end{bmatrix}.$$

That is,

$$\mathbf{C} = \begin{bmatrix} \cos \angle(d_1, e_1) & \cos \angle(d_1, e_2) & \cos \angle(d_1, e_3) \\ \cos \angle(d_2, e_1) & \cos \angle(d_2, e_2) & \cos \angle(d_2, e_3) \\ \cos \angle(d_3, e_1) & \cos \angle(d_3, e_2) & \cos \angle(d_3, e_3) \end{bmatrix}.$$

Now, for the third coordinate system D , the unit vectors have the following components:

$$\vec{e}_1 = [e_{11} \ e_{21} \ e_{31}]^T, \quad \vec{e}_2 = [e_{12} \ e_{22} \ e_{32}]^T, \quad \vec{e}_3 = [e_{13} \ e_{23} \ e_{33}]^T.$$

Similarly,

$$\vec{d}_1 = [d_{11} \ d_{21} \ d_{31}]^T, \quad \vec{d}_2 = [d_{12} \ d_{22} \ d_{32}]^T, \quad \vec{d}_3 = [d_{13} \ d_{23} \ d_{33}]^T.$$

Now combining all data in a single matrix equation, we define

$$\mathbf{C} = \begin{bmatrix} \cos \angle(d_1, e_1) & \cos \angle(d_1, e_2) & \cos \angle(d_1, e_3) \\ \cos \angle(d_2, e_1) & \cos \angle(d_2, e_2) & \cos \angle(d_2, e_3) \\ \cos \angle(d_3, e_1) & \cos \angle(d_3, e_2) & \cos \angle(d_3, e_3) \end{bmatrix}, \quad E = \begin{bmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \end{bmatrix}$$

The cosine matrix is

$$C^T = \begin{bmatrix} \cos \alpha_{11} & \cos \alpha_{12} & \cos \alpha_{13} \\ \cos \alpha_{21} & \cos \alpha_{22} & \cos \alpha_{23} \\ \cos \alpha_{31} & \cos \alpha_{32} & \cos \alpha_{33} \end{bmatrix}.$$

Hence the fundamental transformation equation:

$$\boxed{D = E C^T}$$

This equation is used to transform vectors from coordinate system E to coordinate system D .

Similarly, to go from D to E , follow the same procedure:

$$\boxed{E = D C}$$

From the question, the given state vectors in ECI frame,

$$\mathbf{r} = [-3501.13 \ 15954.16 \ -9521.59]^T \text{ km}, \quad \mathbf{v} = [-7.7387 \ 4.9198 \ -1.5520]^T \text{ km/s}.$$

The orbital frame (RSW frame) is defined as: (i) Radial direction

$$\hat{\mathbf{R}} = \frac{\mathbf{r}}{\|\mathbf{r}\|}.$$

(ii) Orbital normal

$$\hat{\mathbf{W}} = \frac{\mathbf{r} \times \mathbf{v}}{\|\mathbf{r} \times \mathbf{v}\|}.$$

(iii) Along-track direction

$$\hat{\mathbf{S}} = \hat{\mathbf{W}} \times \hat{\mathbf{R}}.$$

Hence, transforming a vector from the orbital frame O to the inertial frame I using the DCM:

$$[IO] = [\hat{\mathbf{R}} \quad \hat{\mathbf{S}} \quad \hat{\mathbf{W}}].$$

RESULT:

$$\mathbf{R}_{\hat{h}} = [-0.1852 \quad 0.8438 \quad -0.5036]^T$$

$$\mathbf{S}_{\hat{h}} = [-0.3272 \quad -0.5362 \quad -0.7781]^T$$

$$\mathbf{W}_{\hat{h}} = [-0.9266 \quad 0.0207 \quad 0.3754]^T$$

Inverse of the DCM:

From the properties of DCMs,

$$[IJ]^{-1} = [IJ]^T,$$

because all DCMs are orthogonal matrices.

Thus,

$$[OI]^{-1} = [IO]^T.$$

RESULT:

$$[IO] = \begin{bmatrix} -0.1852 & -0.3272 & -0.9266 \\ 0.8438 & -0.5362 & 0.0207 \\ -0.5036 & -0.7781 & 0.3754 \end{bmatrix}$$

$$[IO] = \begin{bmatrix} -0.1852 & -0.3272 & -0.9266 \\ 0.8438 & -0.5362 & 0.0207 \\ -0.5036 & -0.7781 & 0.3754 \end{bmatrix}$$

$$[IO]^T = \begin{bmatrix} -0.1852 & 0.8438 & -0.5036 \\ -0.3272 & -0.5362 & -0.7781 \\ -0.9266 & 0.0207 & 0.3754 \end{bmatrix}$$

Verification of the DCM Relation: The goal is to verify that

$$[BI] = [BO][OI]$$

given the matrices $[BO]$ and $[BI]$, where these matrices transform vectors from the ECI frame to the Body frame B .

Given:

$$[BO] = \begin{bmatrix} 0.9397 & 0.1710 & -0.2962 \\ 0.0594 & 0.7713 & 0.6337 \\ 0.3368 & -0.6131 & 0.7146 \end{bmatrix}$$

$$[BI] = \begin{bmatrix} 0.0445 & 0.6951 & -0.7175 \\ -0.8506 & -0.3503 & -0.3921 \\ -0.5239 & 0.6277 & 0.5757 \end{bmatrix}$$

Thus, the relation to check is:

$$[BI] = [BO][OI].$$

RESULT:

$$[BI] = \begin{bmatrix} 0.0445 & 0.6951 & -0.7175 \\ -0.8506 & -0.3503 & -0.3922 \\ -0.5239 & 0.6277 & 0.5757 \end{bmatrix}$$

0.7 Week 7: Attitude Representations

7.1 Euler's Principal Rotation Theorem

According to the definition mentioned in Week7 lecture notes, this theorem states:

“A rigid body or coordinate reference frame can be brought from an arbitrary initial orientation to an arbitrary final orientation by a single rigid rotation through a principal angle Φ about the principal rotation axis $\hat{e} = v$. The principal rotation axis is fixed in both the initial and final orientations.”

Φ = Euler principal rotation angle, $\hat{e}$ = Euler principal rotation axis.

Let B and N be two reference frames. And B can be obtained from N by rotating about the axis $\hat{e}$ (the principal axis) through a quantity Φ (the principal angle). For the given matrix $[BI]$:

$$[BI] = \begin{bmatrix} 0.0445 & 0.6951 & -0.7175 \\ -0.8506 & -0.3503 & -0.3922 \\ -0.5239 & 0.6277 & 0.5757 \end{bmatrix}$$

The Euler Principal angle can be calculated using the formula mentioned in Week7 lecture notes:

$$\Phi = \cos^{-1} \left(\frac{\text{Tr}([BI]) - 1}{2} \right)$$

and Euler's Principal axis:

$$e_1 = \frac{b_{32} - b_{23}}{2 \sin \Phi}, \quad e_2 = \frac{b_{13} - b_{31}}{2 \sin \Phi}, \quad e_3 = \frac{b_{21} - b_{12}}{2 \sin \Phi}.$$

Therefore,

$$\mathbf{e} = \begin{bmatrix} e_1 & e_2 & e_3 \end{bmatrix}^T = \frac{1}{2 \sin \Phi} \begin{bmatrix} b_{32} - b_{23} \\ b_{13} - b_{31} \\ b_{21} - b_{12} \end{bmatrix}$$

Note:

$$\cos \Phi = [-1, 1], \quad \sin \Phi = [-1, 1].$$

RESULT:

- Axis-angle: $\phi = 1.94448284$ rad (111.410660 deg)
- Axis

$$\mathbf{e} = [0.54769776 \quad 0.10397557 \quad 0.83013967] \quad (\|\mathbf{e}\| = 0.99995782)$$

7.2 Quaternions

They are used for attitude parametrization. They describe the orientation of the B frame relative to the I frame, we can use a four-dimensional vector,

$$\beta_{B/I} = [\beta_0 \quad \beta_1 \quad \beta_2 \quad \beta_3]^T$$

$\beta_0, \beta_1, \beta_2, \beta_3$ are Quaternion Elements or Euler Parameters.

Quaternions expressed in terms of principal rotation Φ and principal axis $\mathbf{e} = (e_1, e_2, e_3)^T$:

$$\beta_0 = \cos \frac{\Phi}{2}, \quad \beta_1 = e_1 \sin \frac{\Phi}{2}, \quad \beta_2 = e_2 \sin \frac{\Phi}{2}, \quad \beta_3 = e_3 \sin \frac{\Phi}{2}.$$

Thus quaternion vector:

$$\boldsymbol{\beta} = [\beta_0 \quad \beta_1 \quad \beta_2 \quad \beta_3]^T = \begin{bmatrix} \cos(\Phi/2) \\ \mathbf{e} \sin(\Phi/2) \end{bmatrix}.$$

Norm of quaternion:

$$\|\boldsymbol{\beta}\| = \sqrt{\beta_0^2 + \beta_1^2 + \beta_2^2 + \beta_3^2}.$$

A Quaternion rotation will be valid if:

$$\beta_0^2 + \beta_1^2 + \beta_2^2 + \beta_3^2 = 1.$$

RESULT:

(i) Quaternion: $\mathbf{q} = [q_0 \ q_1 \ q_2 \ q_3] = [0.56344920 \ 0.45248090 \ 0.08589949 \ 0.68582048]$

(ii) Quaternion norm: $\|\mathbf{q}\| = 0.999971207948$

Renormalized quaternion.

7.3 Euler angles (3-1-3 sequence) and rotation matrices

The orientation of the B frame relative to another frame, for eg. I, can be described using three successive rotation about single axes: $[BI] = R_3(\gamma)R_1(\beta)R_3(\alpha)$ This is a 3-1-3 sequence of Euler angles. Using equation 4.32 and 4.34 from Orbital Mechanics,

$$(R_3(\gamma)R_1(\beta)R_3(\alpha))^T = \begin{bmatrix} \cos \gamma \cos \alpha - \sin \gamma \cos \beta \sin \alpha & -\sin \gamma \cos \alpha - \cos \gamma \cos \beta \sin \alpha & \sin \beta \sin \alpha \\ \cos \gamma \sin \alpha + \sin \gamma \cos \beta \cos \alpha & -\sin \gamma \sin \alpha + \cos \gamma \cos \beta \cos \alpha & -\sin \beta \cos \alpha \\ \sin \gamma \sin \beta & \cos \gamma \sin \beta & \cos \beta \end{bmatrix}$$

The kinematic differential equation for the 3-1-3 sequence of Euler angles is given as:

$$\dot{\Theta} = \mathbf{B}(\Theta) \boldsymbol{\omega}$$

Solving for

$$(\dot{\alpha}, \dot{\beta}, \dot{\gamma}) : \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \frac{\sin \gamma}{\sin \beta} & \frac{\cos \gamma \sin \beta}{\sin \beta} & 0 \\ \cos \gamma & -\sin \gamma & 0 \\ -\frac{\sin \gamma \cos \beta}{\sin \beta} & -\frac{\cos \gamma \cos \beta}{\sin \beta} & 1 \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$$\begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix} = \frac{1}{\sin \beta} \begin{bmatrix} \sin \gamma & \cos \gamma & 0 \\ \cos \gamma \sin \beta & -\sin \gamma \sin \beta & 0 \\ -\sin \gamma \cos \beta & -\cos \gamma \cos \beta & 1 \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

Euler Variable Corresponding to Angles

$$\Theta_1 = \alpha, \quad \Theta_2 = \beta, \quad \Theta_3 = \gamma$$

Expression for $\mathbf{B}(\Theta)$ using Euler handouts

$$\mathbf{B}(\Theta) = \frac{1}{\sin \Theta_2} \begin{bmatrix} \Theta_3 & 0 & 0 \\ \sin \Theta_3 & \Theta_2 \cos \Theta_3 & -\sin \Theta_2 \cos \Theta_3 \\ -\Theta_2 \cos \Theta_3 & \Theta_2 \sin \Theta_3 & \sin \Theta_2 \end{bmatrix}$$

RESULT:

$$\mathbf{B}(\theta) = \begin{bmatrix} -0.7837 & 0.9389 & 0 \\ 0.7677 & 0.6408 & 0 \\ 0.4512 & -0.5405 & 1.0000 \end{bmatrix}$$

0.8 Week 8:

$\theta_1 = \alpha; \theta_2 = \beta; \theta_3 = \gamma$ Angular velocity of the spacecraft in B -frame coordinates.

$$\boldsymbol{\omega}_{B/I} = [\omega_1 \quad \omega_2 \quad \omega_3]^T = [-4.3168 \times 10^{-5} \quad 9.2423 \times 10^{-5} \quad 1.04422 \times 10^{-4}]^T \text{ (rad/s)}$$

Differential Equation for $\dot{\boldsymbol{\omega}}$

Let $\vec{\omega}$ be the angular velocity vector of the principal axes frame B as seen from the inertial frame I :

$$\vec{\omega} = \boldsymbol{\omega}_{B/I}$$

Applying the transport theorem gives:

$$\dot{\boldsymbol{\omega}} = \left(\frac{d\boldsymbol{\omega}}{dt} \right)_B + \boldsymbol{\omega} \times \vec{\omega} = \left(\frac{d\boldsymbol{\omega}}{dt} \right)_B$$

$$\dot{\boldsymbol{\omega}} = \dot{\omega}_1 \hat{\mathbf{b}}_1 + \dot{\omega}_2 \hat{\mathbf{b}}_2 + \dot{\omega}_3 \hat{\mathbf{b}}_3$$

Applying the transport theorem on the angular momentum vector:

$$\mathbf{H} = [I] \boldsymbol{\omega}$$

$$\dot{\mathbf{H}} = [I] \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times ([I] \boldsymbol{\omega})$$

$\mathbf{L}$ = total external torque

In the case of Torque-Free Motion, that is, no external torques are acting on the spacecraft, then according to the dynamics of a rigid body,

$$\dot{\mathbf{H}} = \mathbf{L} = 0$$

$$[I] \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times ([I] \boldsymbol{\omega}) = 0$$

Solving for $\dot{\boldsymbol{\omega}}$:

$$\dot{\boldsymbol{\omega}} = -[I]^{-1} (\boldsymbol{\omega} \times ([I] \boldsymbol{\omega}))$$

Rewriting Euler's Equations of Motion

Let principal moments of inertia be I_1, I_2, I_3 . Then:

$$[\dot{\omega}_1 \quad \dot{\omega}_2 \quad \dot{\omega}_3]^T = \left[\frac{(I_2 - I_3)}{I_1} \omega_2 \omega_3 \quad \frac{(I_3 - I_1)}{I_2} \omega_1 \omega_3 \quad \frac{(I_1 - I_2)}{I_3} \omega_1 \omega_2 \right]^T$$

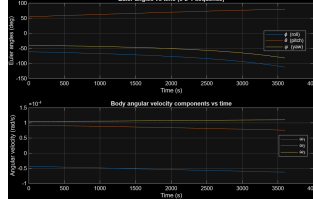


Figure 6: 8.3 Euler Angle and angular velocity

ω = angular velocity vector, $\dot{\omega}$ = time derivative of angular velocity

$\ddot{\omega}$ = angular velocity vector = time derivative of angular velocity $[I]$ = MOI matrix of spacecraft, I_1, I_2, I_3 = principal moments of inertia

State Vector Formulation

$$X = [\Theta^T \ \omega^T]^T$$

$$X = [\Theta_1 \ \Theta_2 \ \Theta_3 \ \omega_1 \ \omega_2 \ \omega_3]^T = [\alpha \ \beta \ \gamma \ \omega_1 \ \omega_2 \ \omega_3]^T$$

Using the 3 – 1 – 3 sequence of rotation matrices:

$$T(\Theta_1, \Theta_2, \Theta_3) = \begin{bmatrix} 0 & -\sin \Theta_1 & \cos \Theta_1 \sin \Theta_2 \\ 1 & \cos \Theta_1 & \sin \Theta_1 \sin \Theta_2 \\ 0 & 0 & \cos \Theta_2 \end{bmatrix}$$

$$\omega = T \dot{\Theta} \Rightarrow \dot{\Theta} = T^{-1} \omega$$

and

$$\dot{\omega} = [I]^{-1} (-\omega \times ([I]\omega))$$

0.8.1 Singularities

Plot

The kinematic differential equation uses $\tan \theta$ and $1/\cos \theta$. For a singularity to occur, $\cos \theta = 0$, that is, $\theta = \pm 90^\circ$. In this case, the initial value of $\theta = 0.9578$ rad (54.9°), which is far from $\pm 90^\circ$. Hence, for these angular velocities, θ will not rapidly change enough to reach $\pm 90^\circ$ in 3600 s. Therefore, the Euler representation will approach singularity.

0.9 Declaration of Originality:

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